80 GHz Horn Antenna with PTFE Lens for Radar Level Sensor

Waveguide horn and screw-on lens for the 60 m version of the 80 GHz level radar board

Overview

An aluminium waveguide horn and a PTFE lens cap made for the 60 m version of the Kymopsis 80 GHz radar level sensor board. The horn bolts to the board over its waveguide launch with three M2 screws; the lens screws onto the M14 × 1.5 thread and seals the horn mouth. Sold as a set, or as a spare horn or spare lens.

Applications

- Spare or replacement antenna for the 60 m 80 GHz level radar board
- Narrow-beam level and distance measurement in tanks, silos, and wells
- Lens and radome experiments at 76–81 GHz

Specifications

Frequency	76–81 GHz (for the 80 GHz level radar board, 60 m version)
Horn	Aluminium; 36.5 mm long; Ø27 mm flange, 2 mm thick; 3 × M2 threaded holes and 3 × Ø3.5 mm holes
Lens thread	M14 × 1.5
Horn mouth	Ø7.8 mm
Waveguide bore	Ø2.7 mm round
Lens	PTFE, screws onto the horn
Gain and beamwidth	Not specified by the maker for the horn on its own; the board datasheet gives range and accuracy with this antenna fitted

Options (Part)

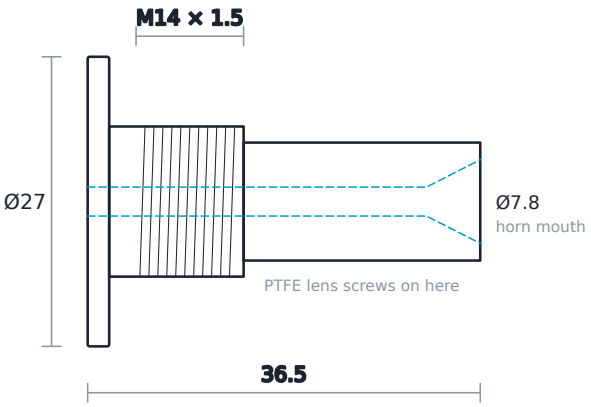
Horn + lens · SKU KYM-LVL80-HORN-SET

Horn only · SKU KYM-LVL80-HORN-H

Lens only · SKU KYM-LVL80-HORN-L

Values are typical or maximum figures stated in the manufacturer's datasheet, not a Kymopsis lab report. Ask us for the manufacturer's test data where one exists.

Outline



80 GHz level radar horn, M14 × 1.5 lens thread

Outline drawing · dimensions in mm · body diameters schematic

Notes

- Fits the 60 m version of the Kymopsis 80 GHz radar level sensor board; it is not a general-purpose waveguide horn with a standard flange.
- Fit the lens for long-range use; keep the lens surface clean and free of condensate.

Passive RF component for laboratory, development and integration use. It does not transmit on its own and carries no FCC, CE or other equipment certification claim.

Kymopsis · kymopsis.com · support@kymopsis.com. Specifications may change without notice. Custom lengths, connectors and bands: kymopsis.com/b2b/